

Fig. 2: Sensor circuit

level never goes below a particular level and no modification in the water tank is required.

Fig. 1 shows the circuit of the water-pump controller. The circuit uses an LDR-white LEDs assembly to sense the water level. It forms a triggering switch to energise the relay for controlling the pump. The LDR-LEDs assembly (shown in Fig. 2) is fixed on the inner side of the cap of the water tank without making contact with water. The light reflected from the water tank is used to control the resistance of LDR1.

When the water level is high enough, light from the white LEDs (LED1 through LED3) reflects to fall on LDR1. This reduces the resistance of LDR1, increasing the voltage at the non-inverting input (pin 3) of IC1. IC1 is used in the circuit as a voltage comparator. Resistors R4 and R5 form a potential divider to fix half of supply voltage to the inverting input of IC1.

Normally, when the water tank is full, LDR1 gets more of reflected light because the distance between the water

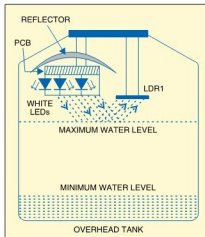


Fig. 3: Sensor assembly

level and the face of LDR1 is minimal. When white light falls on LDR1, the voltage at the non-inverting input (pin 3) of IC1 increases and its output goes high. This high output makes pnp transistor T1 non-conducting and the relay remains de-energised. LED1 also remains 'off.' Since the water-pump power supply is connected to the normally-open (N/O) contacts of relay RL1, pumping is stopped.

When water level falls, the amount of light reflected to LDR1 decreases and its resistance increases. This reduces the voltage at pin 3 of IC1 and its output goes low. This low output from IC1 makes transistor T1 conduct. Relay RL1 energises to close the N/O contacts and the motor starts pumping water. LED1 glows to indicate the

pumping of water.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Solder the white LEDs-LDR1 assembly on a separate PCB and use a separate power supply for it. Mount LEDs behind the LDR. Otherwise, light from the LEDs will affect the working of the circuit. Connect LDR1 to the main circuit board at 'A' and 'B' points.

Fix the LEDs-LDR1 assembly on the inner side of the water-tank cap as shown in Fig. 3. Orient the LEDs and the LDR such that when the water tank is full, the light emitted from the LEDs and reflected from the water surface falls directly on LDR1. The distance between the upper level of water and the LEDs-LDR setup should be minimal, ensuring that water doesn't touch LDR1. Otherwise, the circuit will not function properly. By using more white LEDs, this distance can be increased. Cover the LDR with a black tube to increase its sensitivity.

You can fix the main unit at a convenient place and connect it to the LEDs-LDR assembly through wire. Select the relay according to the horsepower (HP) of the water pump. After arranging the setup (with maximum water in the tank), adjust VR1 until LED1 stops glowing. In this state, the relay should de-energise. When the water level decreases, the relay automatically energises to connect mains to the motor and it starts pumping water. ●

THREE-COMPONENT FLASHER

■ T.A. BABU

Since this flasher system uses only three components, it is relatively easy to build and install. It can be used for signal flashing, hazard warning and alternate flashing.

At the heart of this circuit is a single-pole double-through (SPDT) contacts relay (see Fig. 1). It is a 12V, 400-ohm, single-changeover (1C/O), PCB-mounted relay with contacts cur-

rent rating of 5A.

The working of the circuit is very simple. Initially, bulk capacitor C1 charges through resistor R1 and the bulb filament. When capacitor C1 charges, relay RL1 energises and

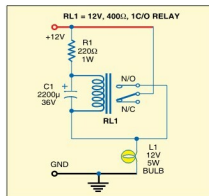


Fig. 1: Simple flasher circuit